

YUSUF YA'U

Software Engineering Student | Machine Learning for Secure & Intelligent Systems
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PROFILE

Software Engineering student with a strong foundation in machine learning and applied AI, combined with hands-on experience building and securing networked, data-driven systems. I have designed and deployed end-to-end AI-driven systems, including a retrieval-augmented generation (RAG) virtual assistant serving live production traffic, and I am developing a CNN-based deep learning tool for automated image and video quality assessment. Through an independent audit of a national digital identity system, I gained practical experience analyzing system security, trustworthiness, and failure modes — work closely aligned with machine-learning-based intrusion and anomaly detection. My research interest centers on applying machine learning and optimization techniques to build reliable, efficient, and secure networked and AI-driven software systems, including IoT and distributed/cloud environments. Currently completing my final semester B.Sc. in Software Engineering at Federal University Dutse with a CGPA of 4.54/5.00 (First Class), I seek to pursue a Master's degree where I can build on this foundation under strong research supervision in machine learning for secure and efficient computing systems.

EDUCATION

Federal University Dutse (FUD), Nigeria

2022 – September 2026

B.Sc. Software Engineering • Matric No: FCP/CSE/22/1053 • CGPA: 4.54/5.00 (First Class)

Relevant Coursework: Artificial Intelligence, Modelling and Simulation, Database Design & Management, Software Engineering, Data Structures & Algorithms, Computer Networks.

RESEARCH INTERESTS

- Machine learning and deep learning for network, IoT, and cloud/distributed systems
- Machine-learning-based security: intrusion, anomaly, and phishing/attack detection
- Robustness, reliability, and trustworthiness of ML-driven and AI/LLM systems
- Machine learning and deep learning for image and video processing
- AI4SE and SE4AI: applying software engineering techniques to validate AI/LLM-driven systems

KEY PROJECTS

AI-Based Image Quality Assessment Tool

Ongoing

Developing a deep learning tool to automatically assess image and video quality by detecting blur, noise, and compression artifacts. Implementing the system in Python using OpenCV and PyTorch/TensorFlow, with CNN architectures such as ResNet for feature extraction. The project provides practical experience in image preprocessing, deep learning, model evaluation, and perception robustness under degraded visual conditions.

3D LiDAR Point Cloud Processing & Visualization

Ongoing

Developing a beginner-level 3D perception project to load, preprocess, visualize, and analyze LiDAR point-cloud data using Python and Open3D. Exploring point-cloud filtering, coordinate transformations, segmentation, and basic object representation as a foundation for future work in LiDAR-based perception.

Federal University Dutse Assistant (FUDA)

Completed

Designed and deployed an AI-powered virtual assistant answering student, applicant, and staff queries. Engineered a retrieval-augmented generation (RAG) pipeline using Python, FastAPI, LangChain, and Meta's LLaMA 3.3-70B via Groq API. Redesigned the retrieval layer into a lightweight keyword-based JSON system after identifying a 512MB RAM limitation, then deployed the system with monitoring and comprehensive technical documentation.

Student Information & Portal Management System

Completed

Building a role-based academic records platform for Government Day Secondary School, Katanga, replacing manual paper-based reporting with a secure digital system. Implementing role-based dashboards using PHP and MySQL for administrators, teachers, and students, together with database design and authentication controls.

Event Management System

Completed

Built a web platform for Dutse Local Government enabling event venue and service owners to register their services and availability. Designed a structured booking workflow matching customers with registered organizers according to event type and date.

EXPERIENCE

StartUp Jigawa SIWES (Student Industrial Work Experience Scheme)

Jul – Dec 2025

Head, Social Media Committee & Software Development Intern

Served as Head of the Social Media Committee and as a Software Development Intern during SIWES. Led the organization's social media presence while contributing to the design and development of “Rent House,” a web application for browsing available rental houses and related listings.

Government Science and Technical College, Birnin Kudu — Mechanical Engineering Workshop

2019 – 2022

Mechanical Engineering Workshop Intern

Gained practical exposure to workshop operations, mechanical tools, equipment handling, basic fabrication, maintenance procedures, and technical problem-solving within a mechanical engineering workshop environment.

VOLUNTEER EXPERIENCE

Junior Science Secondary School, Katanga

2023/2024

Volunteer Computer Studies Teacher

Taught Computer Studies to junior secondary students, building their foundational computing knowledge and practical digital skills.

Hifzul-Gulam Academy School, Katanga

2024/2025

Volunteer Mathematics Teacher & Class Master

Taught Mathematics and served as Class Master, overseeing student discipline, attendance, and classroom coordination. Ran informal remedial sessions to reinforce core mathematical concepts for struggling students.

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** Machine Learning, Deep Learning, CNNs, Model Evaluation, Data Preprocessing, Feature Extraction, Image Quality Assessment
- **Computer Vision & Image/Video Processing:** OpenCV, NumPy, Image/Video Processing, PyTorch (Learning), TensorFlow (Learning)
- **AI/LLM Systems:** Retrieval-Augmented Generation (RAG), Large Language Models (LLMs)
- **3D Perception & Autonomous Systems:** LiDAR Point-Cloud Processing (Learning), Open3D (Learning), Sensor Fusion (Learning), ROS 2 (Learning)
- **Programming:** Python, PHP, JavaScript, C++ (Learning)
- **Frameworks & Tools:** FastAPI, LangChain, Git/GitHub, Linux, Docker (Learning), MATLAB/Simulink (Learning), XAMPP
- **Other:** Academic Writing (APA 7th Edition), Research Proposal Development, Technical Documentation

AWARDS & SCHOLARSHIPS

- Nigerian Federal Government Scholarship 2025/2026 and 2024/2025
- Nasiru Danu Scholarship 2023/2024
- Jigawa State Scholarship 2022/2023
- Best Physics Student, Government Science and Technical College Birnin Kudu 2022/2023

CERTIFICATIONS & TRAINING

- Research for Clinical Innovation, Federal University Dutse 2026
- Introduction to Computer Vision (SoloLearn) *In Progress*
- Introduction to Data Science, Cisco Networking Academy *In Progress*
- CS205: Building with Artificial Intelligence (Saylor Academy) — Grade 88.24 (2025)
- Introduction to Cybersecurity, Cisco Networking Academy *June 2026*
- Machine Learning (SoloLearn) 2023/2024
- Research Methodology, Data Analysis and Publication (International Workshop), Maryam Abacha American University of Nigeria *June 2026*

LANGUAGES

English (Fluent) • Hausa (Native)